QP Code: _____ Reg. No : .....

Name : .....…

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) EXAMINATION

Semester 5: MODEL QUESTION PAPER

Computational Physics: Python

MG5DSEPHY301

Time: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A (Answer any 10 questions)

1. Differentiate between a list and a tuple in Python, with an example of each. *(CO1)*

2. Trace the output of the following code and write the result:

```
fruits = ['mango', 'apple', 'grapes']
fruits.append('kiwi')
fruits.reverse()
print(fruits)
```

(CO1)

3. Differentiate between a set and a dictionary in Python, with a suitable example for each. *(CO1)*

4. Write a Python code snippet using if...elif...else to check whether a given number is positive, negative, or zero. *(CO2)*

5. Write a Python program using a for loop to print numbers from 1 to 5. *(CO2)*

6. What is pickling in Python? Mention the module used and one situation where it is useful. *(CO2)*

7. Write the Python statement(s) to compute the factorial of 5 using the math module. *(CO3)*

8. How do you create a NumPy array from a Python list? Write the code with an example. *(CO3)*

9. Differentiate between array indexing and array slicing in NumPy, with an example. *(CO3)*

10. Write Python code using matplotlib to plot a simple line graph for the points x = [1,2,3,4] and y = [10,20,25,30]. *(CO3)*

11. What is the use of the linspace() function in NumPy? Give an example. *(CO3)*

12. Write Python code to find the maximum value in the NumPy array [3, 7, 2, 9, 4]. *(CO3)*

[2 X 10 = 20]

PART B (Answer any 5 questions)

13. Explain any six list operations in Python with suitable examples (e.g., append, reverse, insert, remove/del, sort, count). *(CO1)*

14. Write a Python program to demonstrate dictionary operations: create a dictionary, add/update a key-value pair, use get(), keys(), values(), items(), and delete an entry using pop(). *(CO1)*

15. Write a Python program using a user-defined function to check whether a given number is prime. Use it to display all prime numbers between 1 and 50. *(CO2)*

16. Explain the basic concept of Object-Oriented Programming in Python. Write a program to create a class 'Student' with attributes name and marks, and a method to display the student's details.*(CO2)*

17. Write a Python program to read data from a CSV file using pandas and display the first five rows. Also write code to write a DataFrame to a new CSV file. *(CO2)*

18. Write a Python program using NumPy to create a 3×3 matrix and perform the following operations: transpose, matrix multiplication, and determinant. *(CO3)*

19. Write a Python program to generate 100 random numbers using NumPy's random module and plot a histogram of the data using matplotlib. *(CO3)*

20. Explain, with a Python program, how to create subplots to display a line plot, a scatter plot, and a bar plot of given data in a single figure, with proper titles and axis labels. *(CO3)*

[6 X 5 = 30]

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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION
Semester 5
PHYSICS OF ATMOSPHERE
(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A — SHORT ANSWER QUESTIONS

Answer any 10 questions. Each question carries 2 marks.

CO1 — 5 questions; CO2 — 4 questions; CO3 — 3 questions

1. What are the major gaseous constituents of the Earth's atmosphere? [K] [CO1]

2. State the hydrostatic equation and explain the physical meaning of each term. [U] [CO1]

3. Define potential temperature. [K] [CO1]

4. Distinguish between weather and climate. [U] [CO1]

5. What are climate indices? Give two examples. [K] [CO1]

6. State the Stefan–Boltzmann law and write its mathematical expression. [K] [CO2]

7. What is meant by radiative equilibrium? [K] [CO2]

8. What is Rayleigh scattering? Mention the condition for Rayleigh scattering in terms of particle size and wavelength. [K] [CO2]

9. Briefly explain Beer's law. [U] [CO2]

10. Define atmospheric vorticity. [K] [CO3]

11. What is Coriolis force? What is its direction? [K] [CO3]

12. What is the difference between geostrophic wind and gradient wind? [U] [CO3]

[10 × 2 = 20 marks]

PART B — SHORT ESSAY QUESTIONS

Answer any 5 questions. Each question carries 6 marks.

CO1 — 3 questions; CO2 — 3 questions; CO3 — 2 questions

13. Write a note on the chemical composition of the atmosphere. [U] [CO1]

14. Explain the hydrologic cycle in detail, describing evaporation, condensation, precipitation, infiltration, runoff and groundwater flow. [U] [CO1]

15. Explain the concept of scale height and discuss its dependence on temperature and molecular weight. [U] [CO1]

16. Explain Planck's radiation law and describe the dependence of spectral radiance on wavelength and temperature. [U] [CO2]

17. Explain Kirchhoff's law of thermal radiation. Show why a good absorber is also a good emitter. [U] [CO2]

18. Explain the greenhouse effect and describe how it increases the surface temperature of the Earth. [U] [CO2]

19. Derive the equation for the geostrophic wind field from the horizontal equations of motion. [A] [CO3]

20. Explain why the atmospheric general circulation can be regarded as a heat engine. [An] [CO3]

[5 × 6 = 30 marks]

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